Relation Algebra

§9. $\frac{R}{S}$

(9a)

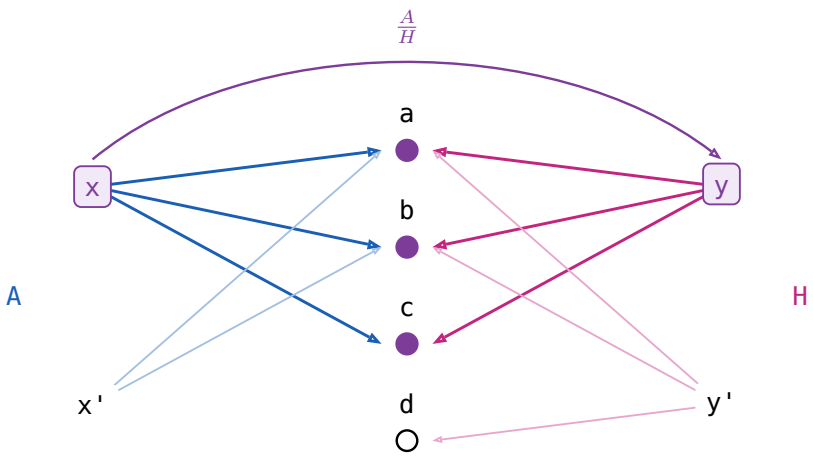
`R /ₛ S ≜ (R / S) ∩ (S / R)°` what `R` sends `x` to and nothing else is what `S` sends `y` to. In `Rel` `x` and `y` has the same image: `∀p. (x R p⇔y S p)`

(9b)

`x (A/H) y` — `x` admires everyone `y` hates
`x ((H/A)°) y` — `x` admires only people `y` hates
`x ((A/H)∩(H/A)°) y` — `x` admires only and all whom `y` hates

`x` admires exactly whom `y` hates: `/` is **all of**, $\frac{R}{S}$ is **only all of**.

(9c)



(9d)

	law	the reading
1	`X ⊑ R /ₛ S ⇔ X S ⊑ R ∧ X° R ⊑ S` `X` sits inside the matching exactly when it carries `S` back inside `R` and, reversed, carries `R` back inside `S`.	`X` may pair `x` with `y` only when `x` admires exactly whom `y` hates. Both halves must typecheck, so the operation is **partial**.
2	`(X /ₛ S)° = S /ₛ X` reversing the matching swaps the two relations.	Matching is symmetric.
3	`(X /ₛ S) (S /ₛ W) ⊑ X /ₛ W` a match from `R` to `S` followed by one from `S` to `W` is a match from `R` to `W`.	And transitive.
4	`(X /ₛ S) S ⊑ X` following the matching by `S` lands inside `R`.	$(\exists y.\, x(\frac{R}{S})y \wedge ySp) \to xRp$ `x only admires whom y hates` $\frac{R}{S}S = \mathrm{Dom}(\frac{R}{S})R$

5	`(X /ₛ X) X = X` matching `R` against itself and then following `R` gives `R` back.	$(\exists y.\, x(\frac{R}{R})y \wedge yRp) \iff xRp$
		`x and y admire the same people`
		`y=x always qualifies (𝟙⊑R%R below)`

6	`𝟙 ⊑ X /ₛ X` every `x` matches itself.	$x(\frac{R}{R})y$ if `x` and `y` admires the same peoples.
7	`(X /ₛ X)² = X /ₛ X` matching twice matches no more pairs than matching once.	So the relation **admires the same people** is an equivalence relation.
8	`Symmetric(X) ⇒ X ⊑ R /ₛ R ⇔ X R ⊑ R` a symmetric `X` sits inside the matching exactly when following it by `R` adds nothing to `R`.	The largest symmetric arrow that leaves `R` alone.
9	`simplePart(R) ≜ R /ₛ 𝟙` matching `R` against the identity keeps the `x` whose image is a single point.	The people who admire exactly one person and nobody else. It equals `R` only when `R` is simple, unlike `R/𝟙=R`.
10	`dom(X /ₛ S) = 𝟙 ∩ ((X / S) (S / X))` the `x` that match something are those the two one-sided divisions take round the loop back to `x`.	Its domain is the **domain of simplicity** of `R`.